P58 A组 Excellent

(9) $2x-5\leq 7$

$\Rightarrow 2x\leq 7+5$

$\Rightarrow 2x\leq 12$

$\Rightarrow x\leq 12/2$

$\Rightarrow x\leq 6$

(10) $x^2-x-6>0$

$\Rightarrow x^2-x-6=0$

Vieta's Formula

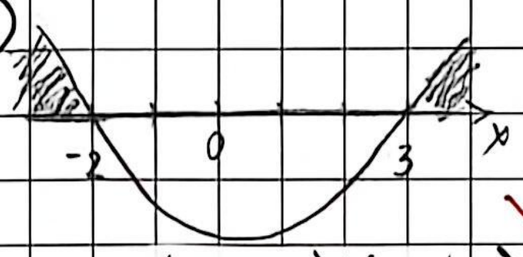
$x_1+x_2=-(-1)$

$x_1x_2=-6$

$x_1=3$

$x_2=-2$

$\Rightarrow x<-2$ 和 $x>3$ $(-\infty;-2)\cup(3;+\infty)$



(11) $x(x-1)(x+2)\leq 0$

$\Rightarrow x_1=0$

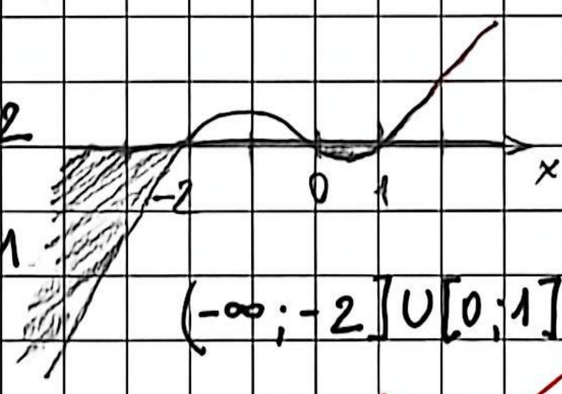
$x_2=1$

$x_3=-2$

$\Rightarrow x\leq -2$

$0\leq x\leq 1$

$(-\infty;-2]\cup[0;1]$



(12) $\frac{x-3}{x+1}<0$

$\Rightarrow x-3=0$; $x_1=3$

$x+1=0$; $x_2=-1$



$-1<x<3$

P59 B组

(6) $\frac{x+1}{x-2}+\frac{x-2}{x+1}=\frac{5}{2}$ $\backslash (x+1)(x-2)$

$\Rightarrow 2(x+1)^2+2(x-2)^2=5(x+1)(x-2)$

$\Rightarrow 2(x^2+2x+1)+2(x^2-4x+4)=5(x^2-2x+x-2)$

$\Rightarrow 2x^2+4x+2+2x^2-8x+8=5x^2-5x-10$

$\Rightarrow 4x^2-4x+10-5x^2+5x+10=0$

$\Rightarrow -x^2+x+20=0$

$\Leftrightarrow x^2-x-20=0$

$\Rightarrow \Delta=(-1)^2-4\cdot 1\cdot(-20)=1+80=81$

$\Rightarrow x_{1,2}=\frac{-(-1)\pm\sqrt{81}}{2\cdot 1}=\frac{1\pm 9}{2}$

$x_1=\frac{1+9}{2}=5$ $x_2=\frac{1-9}{2}=-4$

(7) $2x^2-2x-4\geq 0$

$\Rightarrow 2x^2-2x-4=0 \;/:2$

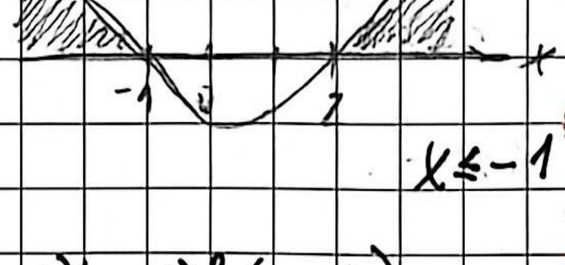
~~$\Delta=(-2)^2-4\cdot 1\cdot(-4)=4+4$~~

$\Rightarrow x^2-x-2=0$

$\Delta=(-1)^2-4\cdot 1\cdot(-2)=9$

$x_1=\frac{1+\sqrt{9}}{2}=\frac{1+3}{2}=\frac{4}{2}=2$

$x_2=\frac{1-\sqrt{9}}{2}=\frac{1-3}{2}=\frac{-2}{2}=-1$



$x\leq -1$ 或 $x\geq 2$

or

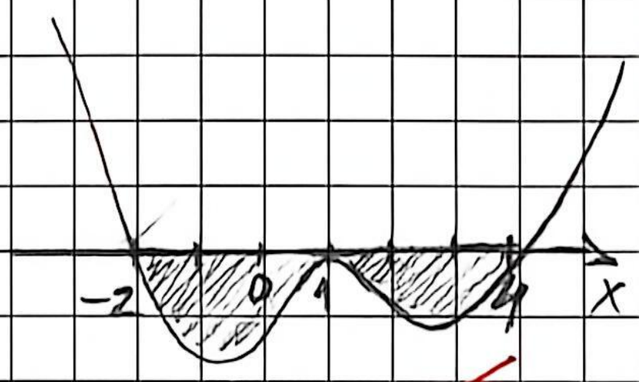
(8) $(x+2)(x-1)^2(x-4)\leq 0$

$\Rightarrow (x+2)(x-1)^2(x-4)=0$

$\Rightarrow x_1=-2$

$x_{2,3}=1$

$x_4=4$

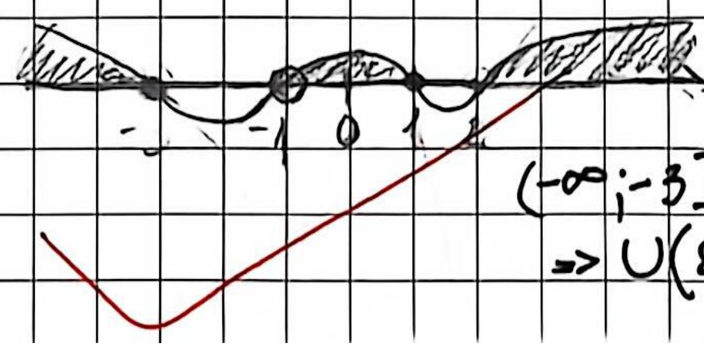


$-2\leq x\leq 4$

(9) $\frac{(x-1)(x+3)}{(x-2)(x+1)}\geq 0$ ① ②

① $\Rightarrow (x-1)(x+3)=0 \Rightarrow x_1=1$; $x_2=-3$

② $\Rightarrow (x-2)(x+1)=0 \Rightarrow x_3=2$; $x_4=-1$



$(-\infty;-3]\cup(-1;1]\cup(2;+\infty)$

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PGO C组

(3) $\frac{x^2-5x+6}{x^2-4} \leq 0$

① $x^2-5x+6=0$

$\Delta = (-5)^2-4\cdot 1\cdot 6 = 25-24=1$

$x_{1,2} = \frac{-(-5)\pm\sqrt{1}}{2\cdot 1} = \frac{5\pm 1}{2}$

$x_1 = 3 \quad x_2 = 2$

$\Rightarrow x^2-5x+6 = (x-3)(x-2)$

② $x^2-4=0$

$x^2-4 = (x-2)(x+2)$

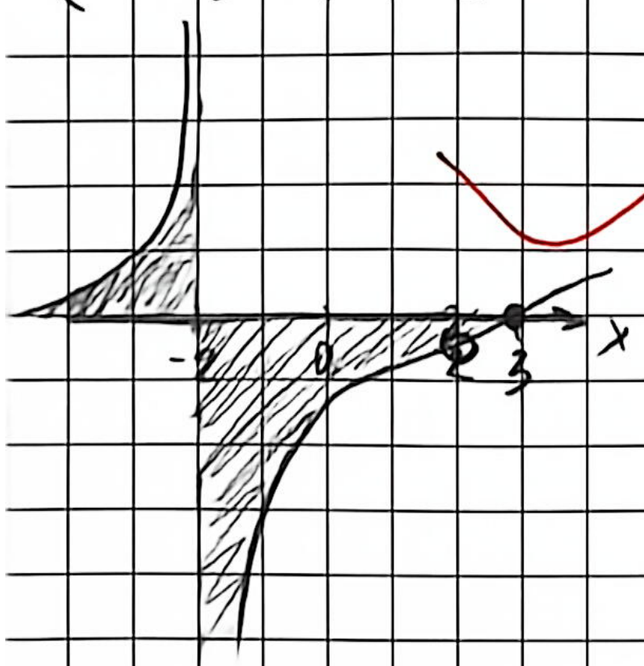
$\frac{(x-2)(x-3)}{(x-2)(x+2)} \leq 0$

$x \neq 2 \; ; \; x \neq -2$

$\Rightarrow \frac{x-3}{x+2} \leq 0 \; ; \; x \neq 2$

~~$-2 < x \leq 2$~~

$(-2;2) \cup (2;3]$



(4) $x^2-2mx+m+2>0$

$a=1 \; ; \; b=-2m \; ; \; c=m+2$

$\Delta = b^2-4ac$

$\Rightarrow \Delta = (-2m)^2-4\cdot 1\cdot (m+2)$

$\Rightarrow \Delta \quad 4m^2-4m-8$

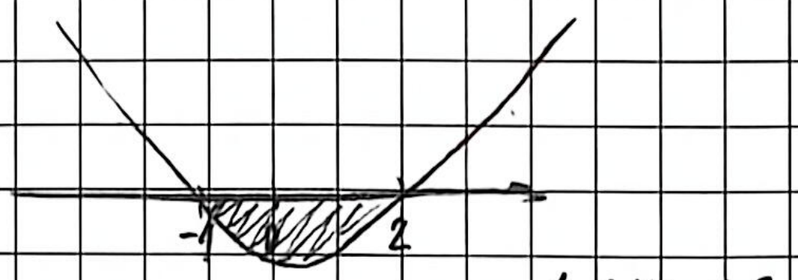
$\Delta < 0 \Rightarrow 4m^2-4m-8<0 \; /:4$

$\Rightarrow m^2-m-2<0$

$\Rightarrow m^2-m-2=0$

$\Rightarrow (m-2)(m+1)=0$

$m_1=2 \; ; \; m_2=-1$



$-1<m<2$

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P.58 A(9)(10)(11)(12)

(9) 解 $2x-5\leq 7$，并写出解集。

$\Rightarrow 2x\leq 7+5$

$\Rightarrow 2x\leq 12$

$\Rightarrow x\leq 6$

$x=(-\infty,6]$

(10) 解 $x^2-x-6>0$

$\Rightarrow (x-3)(x+2)>0$

$\Rightarrow x=3,-2$

- $x<-2$: $(-)(-)=+$
- $-2<x<3=(-)(+)$
- $x>3=(+)(+)=+$

$\Rightarrow x=(-\infty,-2)\cup(3,+\infty)$

(11) 解 $x(x-1)(x+2)\leq 0$

$\Rightarrow x=0,1,-2$

- $x<-2$: $(-)(-)(-)=-$
- $-2<x<0$: $(-)(-)(+)=+$
- $0<x<1$: $(+)(-)(+)=-$
- $x>1$: $(+)(+)(+)=+$

$\Rightarrow x=(-\infty,2]\cup[0,1]$

(12) 解 $\frac{x-3}{x+1}<0$

$\Rightarrow (x-3)(x+1)<0$

$\Rightarrow x=3,-1$

- $x<-1$: $(-)(-)=+$
- $-1<x<3$: $(-)(+)=-$
- $x>3$: $(+)(+)=+$

$\Rightarrow x=(-1,3)$

P.59 B(6)(7)(8)(9)

(6) 解 $\frac{x+7}{x-2}+\frac{x-2}{x+1}=\frac{5}{2}$

$=2(x+1)^2+2(x-2)^2=5(x-2)(x+1)$

$=2(x^2+2x+1)+2(x^2-4x+4)=5(x^2-x-2)$

$=2x^2+4x+2+2x^2-8x+8=5x^2-5x-10$

$=4x^2-8x+10=5x^2-5x-10$

$=x^2-x-20=0$

$=(x-5)(x+4)$

$\Rightarrow x=5,-4$; $x\neq 2,-1$

(7) 解 $2x^2-2x-4\geq 0$

$=x^2-x-2\geq 0$

$=(x-2)(x+1)\geq 0$

$=x=2,-1$

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RELIVE THE TOUCH OF PEN HOLDING

- $x<-1$: $(-)(-)=+$
- $-1<x<2$: $(-)(+)=-$
- $x>2$: $(+)(+)=+$

$\Rightarrow x=(-\infty,-1]\cup[2,\infty)$

(8) 解 $(x+2)(x-1)^2(x-4)\leq 0$

$\Rightarrow x=-2,1,4$

- $x<-2$: $(-)(+)(-)=+$
- $-2<x<1$: $(+)(+)(-)=-$
- $x>4$: $(+)(+)(+)=+$

$\Rightarrow x=[-2,4]$

(9) 解 $\frac{(x-1)(x+3)}{(x-2)(x+1)}\geq 0$

$\Rightarrow (x-1)(x+3)(x-2)(x+1)\geq 0$

$\Rightarrow x=1,-3,2,-1$

- $x<-3$: $(-)(-)(-)(-)=+$
- $-3<x<-1$: $(-)(+)(-)(-)=-$
- $-1<x<1$: $(-)(+)(-)(+)=+$
- $1<x<2$: $(+)(+)(-)(+)=-$
- $x>2$: $(+)(+)(+)(+)=+$

$\Rightarrow x=(-\infty,-3]\cup(-1,1]\cup(2,\infty)$

P.60 (3)(4)

(3) 解不等式 $\frac{x^2-5x+6}{x^2-4}\leq 0$

$=\frac{(x-2)(x-3)}{(x-2)(x+2)}\leq 0$

$=\frac{(x-3)}{(x+2)}\leq 0$

$=(x-3)(x+2)\leq 0$

$\Rightarrow x=3,-2$

- $x<-2$: $(-)(-)=+$
- $-2<x<3$: $(-)(+)=-$
- $x>3$: $(+)(+)=+$

$\Rightarrow x=(-2,2)\cup(2,3]$

(4) 求所有实数 m，使不等式

$x^2-2mx+m+2>0$ 对一切

实数 x 恒成立。

- 二次项系数 $a>0$
- $\Delta<0$

$\Delta=b^2-4ac$

$=(-2m)^2-4(1)(m+2)$

$=4m^2-4m-8$

$\Delta<0$

$4m^2-4m-8<0$

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$m^2 - m - 2 < 0$

$(m-2)(m+1) < 0$

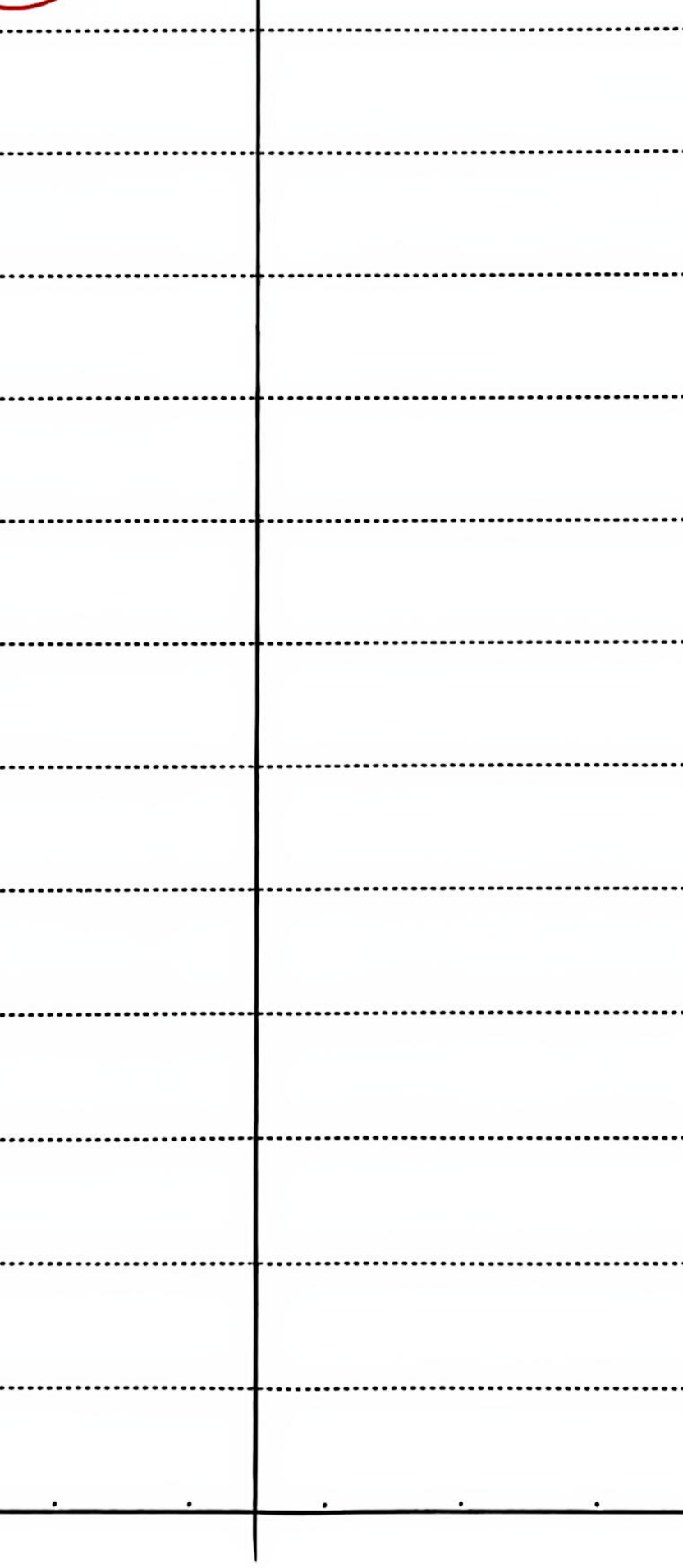
$m = 2 ; -1$

• $m < -1 : (-)(-) : +$

• $-1 < m < 2 : (-)(+) : -$

• $m > 2 : (+)(+) : +$

$= -1 < m < 2$



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Homework Excellent!

P58

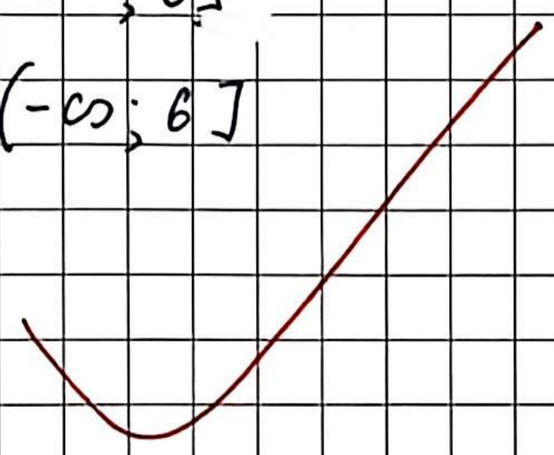
A9) $2x-5 \leq 7$

$2x \leq 12$

$x \leq 6$

$x \in (-\infty; 6]$

$-\infty \to \infty$



Answer: $(-\infty; 6]$

A10) $x^2 - x - 6 > 0$

By Vieta's formulas: $(x-3)(x+2)=0 \Rightarrow$

$x_1 = 3$, $x_2 = -2$

Use the interval method:

For $x < -2$, the expression (+)

For $-2 < x < 3$, the expression (-)

For $x > 3$, the expression (+)

$x < -2$ or $x > 3$

Answer: $(-\infty; -2) \cup (3; +\infty)$

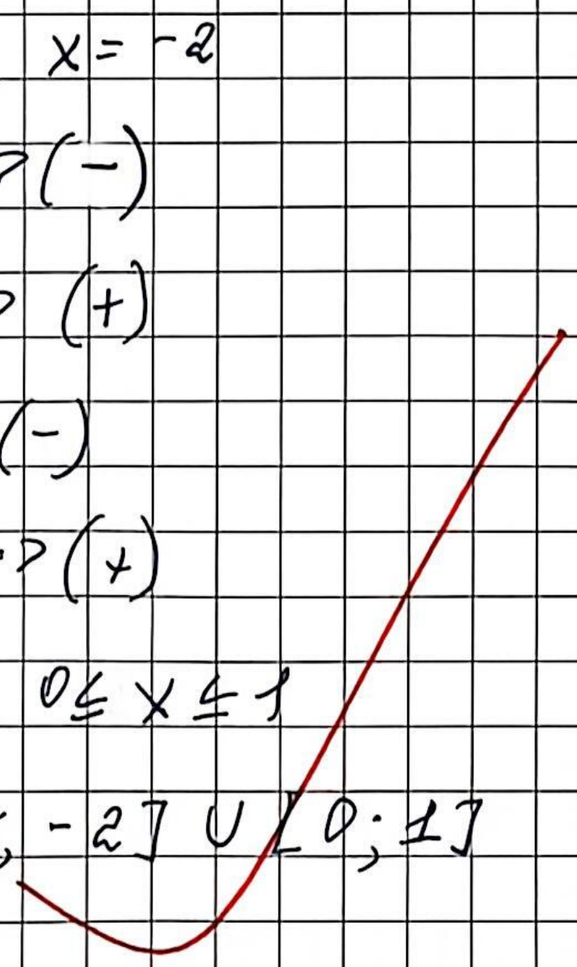
A11) $x(x-1)(x+2) \leq 0$

$x = 0$, $x = 1$, $x = -2$

- $(-\infty; -2) \Rightarrow (-)$
- $[-2; 0] \Rightarrow (+)$
- $[0; 1] \Rightarrow (-)$
- $[1; +\infty) \Rightarrow (+)$

$x \leq -2$ or $0 \leq x \leq 1$

Answer: $(-\infty; -2] \cup [0; 1]$



A12) $\frac{x-3}{x+1} < 0$

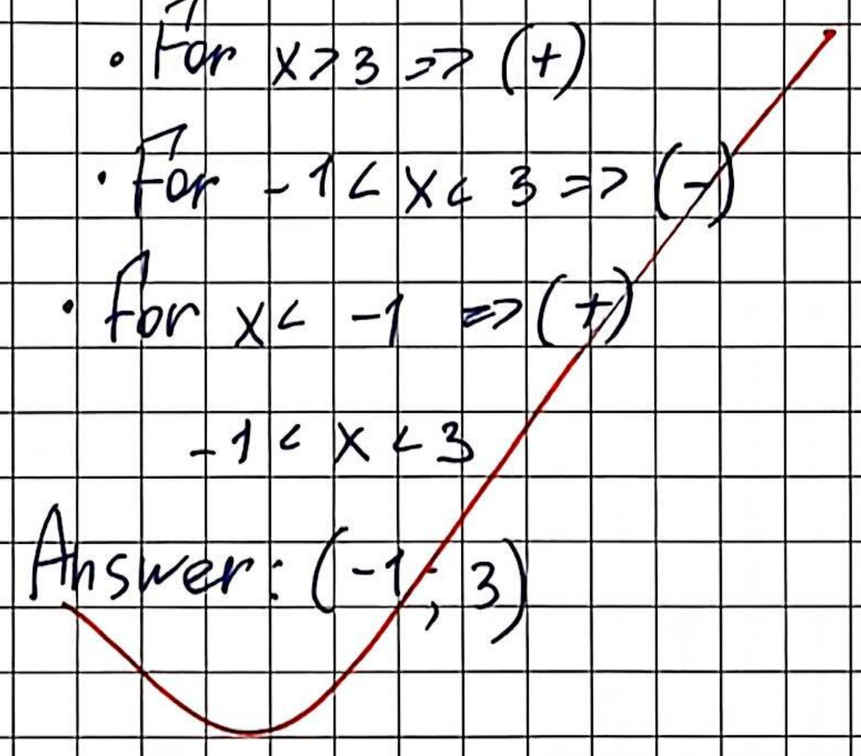
$x - 3 = 0 \Rightarrow x = 3$

$x + 1 \neq 0 \Rightarrow x \neq 1$

- For $x > 3 \Rightarrow (+)$
- For $-1 < x < 3 \Rightarrow (-)$
- for $x < -1 \Rightarrow (+)$

$-1 < x < 3$

Answer: $(-1; 3)$



P59

B6) $\frac{x+1}{x-2} + \frac{x-2}{x+1} = \frac{5}{2}$

Let $t = \frac{x+1}{x-2}$, $t + \frac{1}{t} = \frac{5}{2}$

$2t^2 + 2 = 5t \Rightarrow 2t^2 - 5t + 2 = 0$

$(2t-1)(t-2) = 0 \Rightarrow t = \frac{1}{2}$ or $t = 2$

Case 1: if $t = 2$

$\frac{x+1}{x-2} = 2 \Rightarrow x + 1 = 2(x-2) \Rightarrow x + 1 = 2x - 4 \Rightarrow x =$

Case 2: if $t = \frac{1}{2}$

$\frac{x+1}{x-2} = \frac{1}{2} \Rightarrow 2(x+1) = x - 2 \Rightarrow 2x + 2 = x - 2 \Rightarrow$

$x = -4$

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Answer: $x=5$, $x=-4$

B7) $2x^2-2x-4\geq 0$

$x^2-x-2\geq 0$

$(x-2)(x+1)\geq 0$

$x\leq -1$ or $x\geq 2$

Answer: $(-\infty;-1]\cup[2;+\infty)$

B8) $(x+2)(x-1)^2(x-4)\leq 0$

$(x+2)(x-4)\leq 0$

$x=-2$ and $x=4$ $\Rightarrow$

$-2\leq x\leq 4$

Answer: $[-2;4]$

B9) $\frac{(x-1)(x+3)}{(x-2)(x+1)}\geq 0$

- From numerator: $x=1$, $x=-3$
- From denominator: $x\neq 2$, $x\neq -1$

$x\leq -3$ or $-1<x\leq 1$ or $x>2$

Answer: $(-\infty;-3]\cup(-1;1]\cup(2;+\infty)$

P60

C3) $\frac{x^2-5x+6}{x^2-4}\leq 0$

- Numerator: $x^2-5x+6=(x-2)(x-3)$
- Denominator: $x^2-4=(x-2)(x+2)$

$x-2\neq 0 \Rightarrow x\neq 2$

$x+2\neq 0 \Rightarrow x\neq -2$

$\frac{(x-2)(x-3)}{(x-2)(x+2)}\leq 0 \Rightarrow \frac{x-3}{x+2}\leq 0$

$-2<x\leq 3$

$x\in(-2;2)\cup(2;3]$

Answer: $(-2;2)\cup(2;3]$

C4) $x^2-2mx+m+2>0$

$a=1$, $b=-2m$, $c=m+2$

$D=b^2-4ac=(-2m)^2-4\cdot 1\cdot(m+2)=$

$=4m^2-4m-8$

$4m^2-4m-8<0$

$m^2-m-2<0$

$(m-2)(m+1)<0$

$-1<m<2$

Answer: $-1<m<2$

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Excellent

P 58 A (9)(10)(11)(12), P59 B (6)(7)(8)(9), P60 C (3)(4)

P 58 A (9)(10)(11)(12)

(9). Solve $2x - 5 \leq 7$, and write the solution set

→ $2x - 5 \leq 7$

$2x \leq 7 + 5$

$2x \leq 12 \div 2$

$x \leq 6$ → $(-\infty, 6)$,, × ⇔ $x < 6$,,

→ $(-\infty, 6]$ ✓

(10). Solve $x^2 - x - 6 > 0$

→ $(x - 3)(x + 2) > 0$

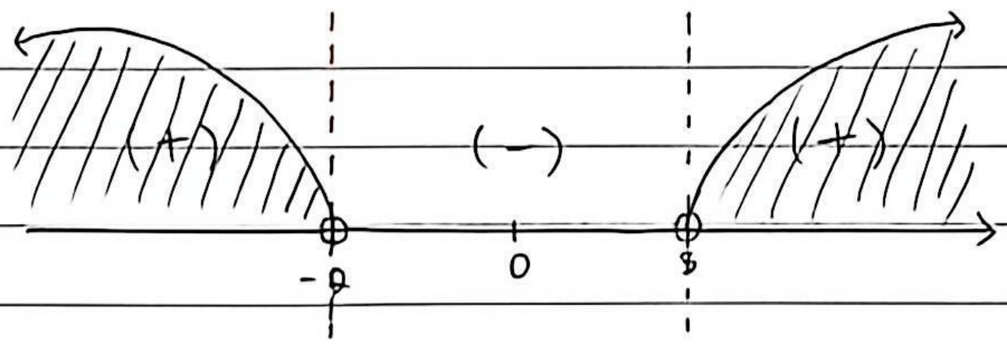
$x - 3 = 0$ ∧ $x + 2 = 0$

$x = 3$,, $x = -2$,,

如 $x = 0$

$0^2 - 0 - 6 = -6$ (−)

$x < -2$ 或 $x > 3$



⇝ $(-\infty, -2) \cup (3, +\infty)$,,

(11) Solve $x(x - 1)(x + 2) \leq 0$

→ $x = 0$,, $x - 1 = 0$, $x + 2 = 0$

$x = 1$ $x = -2$,,

如/if :

$x = -3 \to (-3)(-3-1)(-3+2) = (-)$

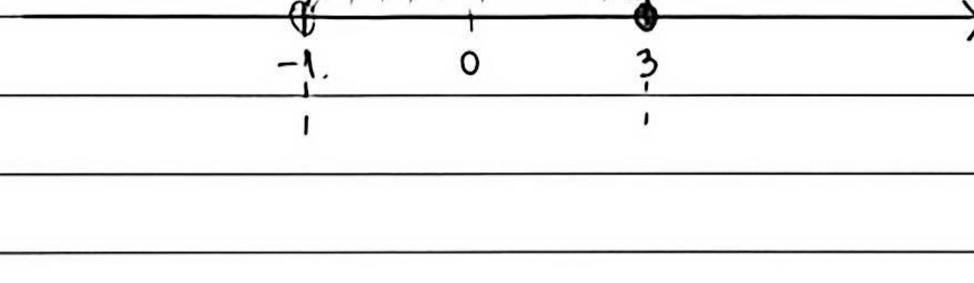
$x = -1 \to (-1)(-1-1)(-1+2) = (+)$

$x = \frac{1}{2} \to (\frac{1}{2})(\frac{1}{2}-1)(\frac{1}{2}+2) = (-)$

$x = 1 \to (2)(2-1)(2+2) = (+)$

Solution $= \{x \mid x \leq -3 \ \vee\ 0 \leq x \leq 1,\ x \in \mathbb{R}\}$,,

$= (-\infty, -2] \cup [0, 1]$,,



(12). Solve $\frac{x-3}{x+1} < 0$

(分母)

→ $x - 3 = 0$ $x + 1 \neq 0$

(分子) $x = 3$,, $x \neq -1$,,

如/if

$x = 0 \to \frac{0-3}{0+1} = \frac{-3}{1} = -3$ (−)

⇝ $(-1, 3]$,,

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P59 B (6)(7)(8)(9)

(6). Solve $\frac{x+1}{x-2} + \frac{x-2}{x+1} = \frac{5}{2}$

→ $x-2 \neq 0$ | $x+1 \neq 0$

$x \neq 2$,, | $x \neq -1$,,

Multiply everything by $(x-2)(x+1)$:

$(x+1)^2 + (x-2)^2 = \frac{5}{2}(x-2)(x+1)$

$= x^2 + 2x + 1 + x^2 - 4x + 4 = \frac{5}{2}(x^2 - x - 2)$

$= 2x^2 - 2x + 5 = \frac{5}{2}x^2 - \frac{5}{2}x - 5$

Multiple (×) by 2 :

$= 4x^2 - 4x + 10 = 5x^2 - 5x - 10$

$= x^2 - x - 20 = 0$

$\rightarrow (x-5)(x+4) = 0$

$x - 5 = 0$ ∨ $x + 4 = 0$ → $x = 5$ 或 $x = -4$,,

$x = 5$,, $x = -4$,, : $\{-4, 5\}$,,

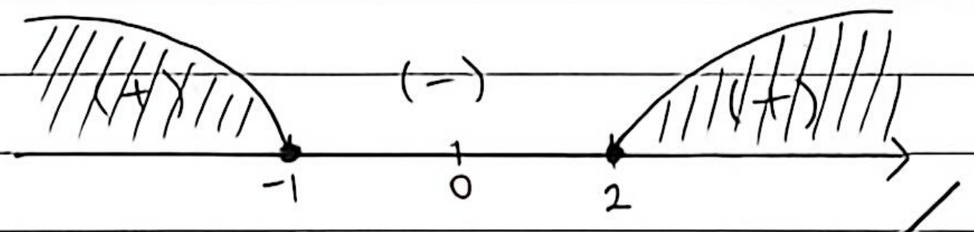
(7). Solve $2x^2 - 2x - 4 \geq 0$

→ $2(x^2 - x - 2) \geqslant 0$

$2(x+1)(x-2) \geqslant 0$

$x + 1 \geqslant 0$ ∨ $x - 2 \geqslant 0$

$x \leqslant -1$,, $x \geqslant 2$,,



→ $x \leqslant -1$ 或 $x \geqslant 2$,,

$= (-\infty, -1] \cup [2, +\infty)$,,

如/if :

$x = 0 \rightarrow 2(0+1)(0-2) = 2(1)(-2) = \underset{-4}{(-2)}(-)$

(8). Solve $(x+2)(x-1)^2(x-4) \leq 0$

→ $x + 2 = 0$ | $(x-1)^2 = 0$ | $x - 4 = 0$

$x = -2$,, | $x = 1$,, | $x = 4$,,

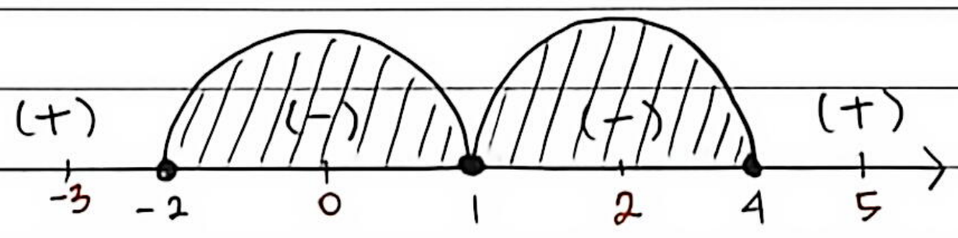
如/if

$x < -3 \rightarrow (-3+2)(-3-1)^2(-3-4) = (+)$

$x = 0 \rightarrow (0+2)(0-1)^2(0-4) = (-)$

$x = 2 \rightarrow (2+2)(2-1)^2(2-4) = (-)$

$x = 5 \rightarrow (5+2)(5-1)^2(5-4) = (+)$



→ Solution : $\{x \mid -2 \leqslant x \leqslant 4, x \in \mathbb{R}\}$,,

$= [-2, 4]$,,

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5g B (9)

(9). Solve $\frac{(u-1)(u+3)}{(u-2)(u+1)} \geq 0$

→ 分子 (Numerator)

$u-1=0$ $\quad u+3=0$

$u=1$„ $\quad u=-3$„

分母 (Denuminator)

$u-2\neq 0$ $\quad u+1=0$

$u\neq 2$„ $\quad u\neq -1$„

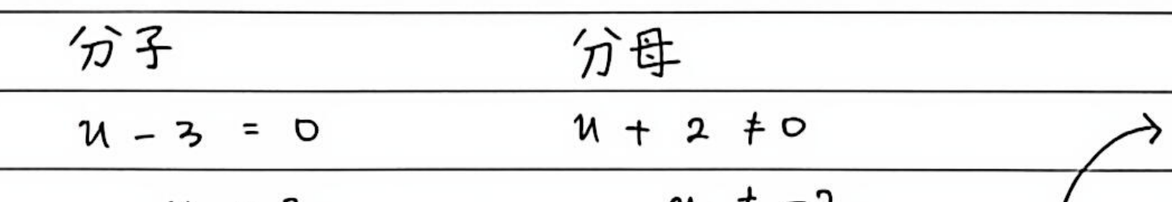
如/if:

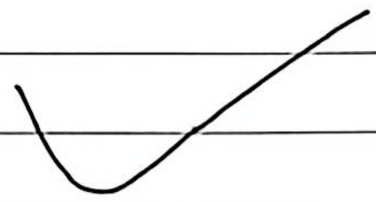
$u=0 \rightarrow \frac{(0-1)(0+3)}{(0-2)(0+1)} = \frac{-3}{-1} = (+)$

$u=-2 \rightarrow \frac{(-2-1)(-2+3)}{(-2-2)(-2+1)} = \frac{-3}{4} = (-)$

→ $u\leq -3$ 或 $-1<u\leq 1$ 或 $u>2$„

$=(-\infty,-3]\cup(-1,1]\cup(2,+\infty)$„





P 60 C (3)(4)

(3). Solve the inequality $\frac{u^2-5u+6}{u^2-4} \leq 0$

→ faktorkan $\frac{(u-2)(u-3)}{(u-2)(u+2)} \leq 0$

$u\neq 2$„

$= \frac{(u-3)}{(u+2)} \leq 0$

如/if:

$u=0 \rightarrow \frac{(0-3)}{(0+2)} = \frac{-3}{2} = (-)$

$u=-3 \rightarrow \frac{(-3)-3)}{(-3)+2)} = \frac{-6}{-1} = (+)$

$u=2,5 \rightarrow \frac{(2,5-3)}{(2,5+2)} = \frac{-0,5}{4,5} = (-)$

分子

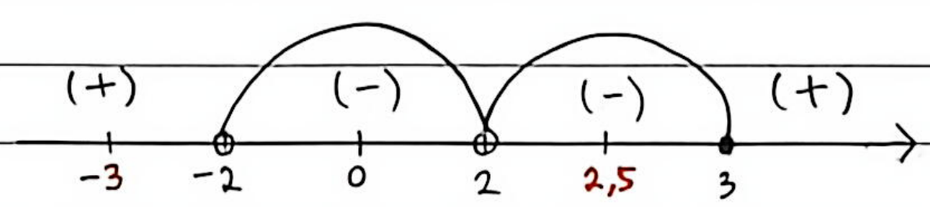
$u-3=0$

$u=3$„

分母

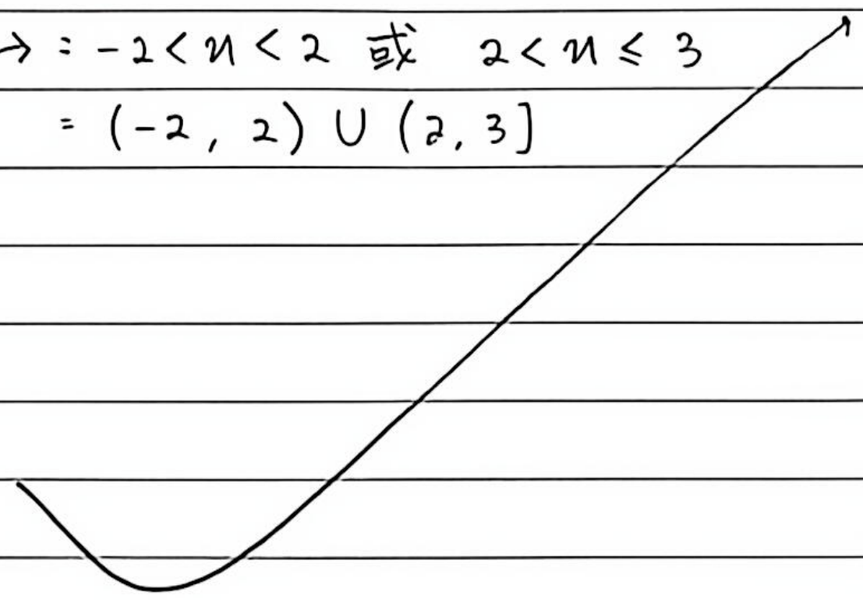
$u+2\neq 0$

$u\neq -2$„



→ $-2<u<2$ 或 $2<u\leq 3$

$=(-2,2)\cup(2,3]$



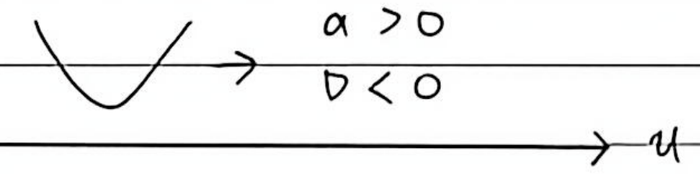
扫描全能王
3亿人都在用的扫描App

(4) Find all real numbers m such that $x^2 - 2mx + m + 2 > 0$ holds for all real numbers x

→ $ax^2 + bx + c$

$x^2 - 2mx + m + 2 > 0$

to get all real numbers $D < 0$ →



∴

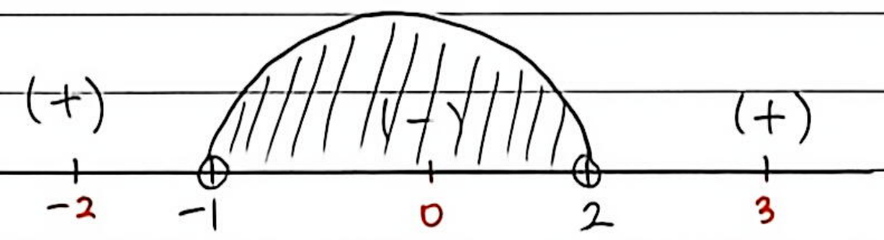
$D < 0$

$b^2 - 4ac < 0$

$(-2m)^2 - 4(1)(m+2) < 0$

$4m^2 - 4m - 8 < 0$ $\times \frac{1}{4}$

$m^2 - m - 2 < 0$	如/if ·
$(m-2)(m+1) < 0$	$x = 0 \rightarrow 0^2 - 0 - 2 = (-)$
$m-2=0$ $\quad m+1=0$	$x = -2 \rightarrow (-2)^2 - (-2) - 2 = (+)$
$m = 2$, $\quad m = -1$,	$x = 3 \rightarrow 3^2 - 3 - 2 = (+)$



→ Solution : $\{m \mid -1 < m < 2, m \in \mathbb{R}\}$,

